Wajahat Mahmood

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Education

CUNY Queens College — Expected Jan 2027
Bachelor of Science in Computer Science — *GPA 3.54/4.0*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Software Engineering, Linear Algebra, Probability

Experience

Software Engineer Intern — Robotics & Test Automation — May 2026 – Present
rf IDEAS, Inc. — RFID/NFC credential reader manufacturer — *Schaumburg, IL*

- Automated a 10-hour manual RFID validation process by building Python control software for a 6-axis robotic arm, enabling unattended overnight test runs and saving ∼20 engineer-hours per week.
- Built a Python calibration system that sweeps 1,600+ reader configurations per device model and measures read success rate at each point, replacing manual parameter tuning with data-driven selection.
- Developed C++ / GTest suites measuring RFID read height to 0.1 mm accuracy on 8 orientations and reader response latency at millisecond resolution, surfacing antenna alignment defects earlier in the hardware test cycle.
- Hardened unattended runs with workspace validation, joint-limit enforcement, autosave checkpointing, and automated fault recovery, roughly halving overnight runs lost to unrecovered faults.
- Streamed live joint telemetry over a custom UDP protocol into ROS 2 and a 3D visualization tool, giving operators real-time visibility into arm state for diagnosing motion faults.

Robotics Software Engineering Researcher — May 2025 – Jul 2025
University of Michigan, Laboratory for Progress — *Ann Arbor, MI*

- Co-authored a paper accepted to **IEEE/RSJ IROS 2026** and presented the research at ABRCMS 2025.
- Implemented C++ teleoperation software for a Trossen Interbotix RX-200 arm, mapping game-controller input via SDL2 to joint-velocity commands through the vendor SDK, enabling reliable remote manipulation of the robot.
- Wrote C++ middleware that bridged the lab's publish-subscribe robot OS to physical hardware, enabling existing lab software to control the arm in real time.
- Diagnosed and eliminated race conditions that intermittently crashed the vendor driver by guarding shared state with mutexes and atomics.

Fellowships

Google Software Engineering Program (G-SWEP) — via Basta — Jun 2026 – Present

CUNY Tech Prep — Software Engineering Fellow — Jul 2026 – Present

Projects

GlanceScribe | *Swift, Node.js, Gemini, ElevenLabs, FFmpeg* — **Best Healthcare Track, Hack Brooklyn 2026**

- Built an AI medical scribe that turns a clinician's first-person video from Meta smart glasses into structured SOAP notes, eliminating manual note-taking during patient encounters.
- Reverse-engineered the glasses' streaming protocol (which has no public video API) to capture live video through a Swift iOS app into a Node.js backend.
- Engineered a multimodal pipeline combining ElevenLabs speech transcription with Gemini vision, using schema-constrained JSON decoding so every generated note parses reliably into the SOAP structure.
- Built an FFmpeg pipeline that scores video segments by duration and model confidence to extract the most clinically relevant clips.

Real-Time Collaborative Code Editor | *Flask, React, WebSockets, Socket.IO, CI/CD*

- Synchronized concurrent edits and user presence across clients over Socket.IO, with server-side session recovery and automatic reconnection so users rejoin mid-session after dropped connections.
- Sandboxed multi-language code execution through the Piston API, isolating untrusted user code from the Flask backend.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, Swift, SQL
Robotics & Systems: ROS 2, multithreading (mutexes, atomics), TCP/UDP sockets, pub/sub architectures, GTest
Web & Frameworks: React, Node.js, Flask, Socket.IO, WebSockets, REST APIs
AI/ML: Gemini API, Claude API, multimodal pipelines, structured (schema-constrained) JSON generation
Tools: Git, GitHub, Linux, CI/CD, FFmpeg, NumPy, Pandas, SDL2